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Neon by Pupil Labs - Live Recording and Other Study Types

This article provide information about using Pupil Labs - Neon Eye Tracking Glasses, including steps for setting up the glasses, collecting data, and importing data into iMotions for Glasses Live Recordings. Importing steps also apply to Glasses Sync and Import Glasses studies (Related article: [Eye Tracking Glasses Study Types](#)).

Requirements for Neon Glasses Recordings

- **Android Phone** - Included with the Neon Glasses (e.g. OnePlus 8T Phone).
- **Neon Companion App** - Pre-installed or installed via [Google Play](#).
 - Neon Companion App version 2.9.3 is supported in iMotions 10.1.13 or newer.
 - Older versions of iMotions are compatible with recordings from Neon Companion App version 2.7.5.
 - Alternative, 'Offline' app ([download link](#)) - Companion app without Pupil Cloud features and log-in requirement (Note: The standard app is recommended, if an Pupil Labs app log-in can be used - please contact iMotions for more information).
- **Compatible phone and app versions** - Described in [Pupil Labs documentation](#).
 - OnePlus 8T requires Android 11 (newer Android versions are incompatible with the Neon Companion App).
- **Pupil Cloud account** - Required log-in for Neon Companion App and upload/processing on Pupil Cloud.

Neon Glasses Signals in iMotions

Please refer to Pupil Labs Neon documentation for detailed information about these data.

- **Timestamp**
- **Pupil***
- **Gaze 2D** (Gaze left and right x/y)
- **Gyroscope** (X, Y, Z)
- **Accelerometer** (X, Y, Z)
- **Behavioral** (Pitch, Yaw, Roll)
- **Other** (Blink, Gaze3d, Fixation)*
- **Eyeball Center** (Left and Right X, Y, Z)*
- **Optical Axis** (Left and Right X, Y, Z)*

*indicates signals/data that require enabled settings in Neon Companion phone app (or Pupil Cloud Processing): **Compute eye state** and **Compute fixations**

Video Camera Recordings, Timestamp, and Gaze 2D are always available for import into iMotions.

Neon Glasses for Live Recordings

Live Recordings will automatically include gaze data and other sensor data (set up and collected in the study).

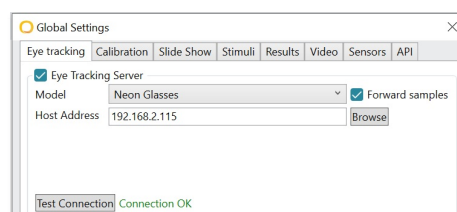
To synchronize glasses video and additional data, files must be exported from the Neon Companion App, transferred onto your computer, and imported into iMotions.

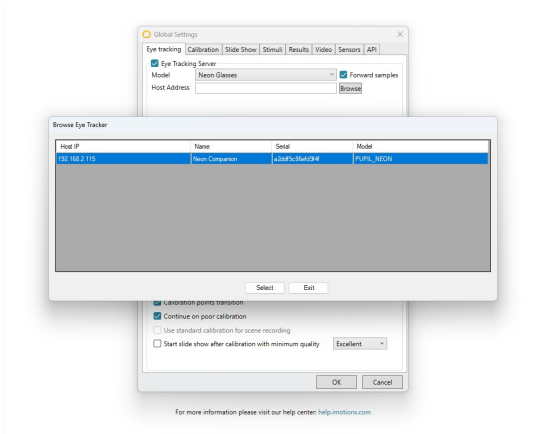
Requirement: For iMotions Studies using Glasses Live Recording and Neon Glasses - your iMotions computer and Neon phone must be connected to the same WiFi network.

- The same WiFi network must be selected in Windows settings on the iMotions computer and in Android settings on the Neon Glasses phone.
- iMotions can then connect to the Neon Glasses via the WiFi network with the following steps (allowing iMotions to start recordings on the Neon Glasses).
- Note: These steps are not necessary for using eye tracking glasses with **Sensor Sync for Glasses** or **Import glasses data** (which are other options for using Eye Tracking Glasses).
- Institutional and Public Wifi networks are usually firewalled to prevent communications directly across devices preventing iMotions from connecting to your phone. You may need to contact your IT dept. Alternatively, it may be easier to set up your own offline wifi network with a router.

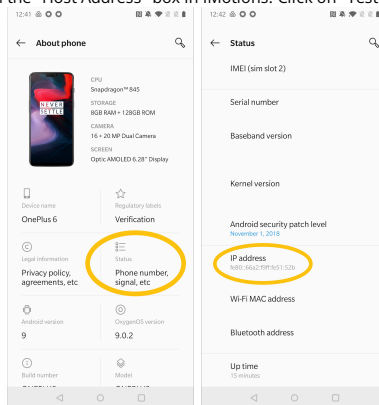
How to Connect Neon Glasses to iMotions

1. Connect the Neon Glasses to the phone with the provided USB C cable.
2. Start the phone and Neon Companion app. If it's your first time using the app you will need to create a new "Wearer".
3. Connect the phone and iMotions computer to the same wifi network.
4. Open iMotions Global Settings (from the Preferences drop-down menu).
5. In the Eye tracking tab, enable/check "Eye Tracking Server". Use the drop-down menu to select "Neon Glasses".
6. Click 'Browse' and select 'Neon Companion' connection (with the IP address, device name and serial number, phone model)
7. Click "Select" when you have found the device. You can test the connection by clicking on the "Test Connection" button. Click on "OK", and then select "OK" when back in Global Settings.





8. If you have multiple Neon glasses running at the same time on the same WiFi, you will need to differentiate your device of choice through the IP address. The phone's IP address can be found on the phone by going to "Settings" (the gear icon), "About phone", and "Status" and then matched to "Host IP" in iMotions.
9. If the IP address of the glasses doesn't appear automatically, you may need to manually enter it. Find the IP address of the phone using the same method as in the above paragraph, and enter this information in the "Host Address" box in iMotions. Click on "Test Connection" to check the connection.



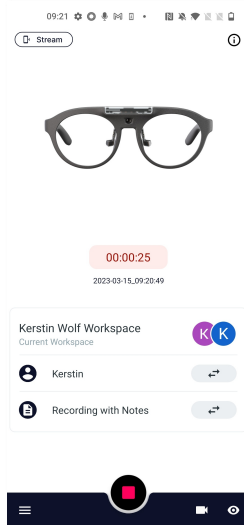
Android phone settings for IP address information

Troubleshooting Note

If any crashes occur with the Neon Companion app are unlikely, please reboot the entire phone and then reconnect to iMotions with these setup steps. This will ensure that the error state has been cleared and does not impact the connection to iMotions (since error states in the Companion App are not detected in iMotions).

Collecting Data with Live Recording

1. Create a study with iMotions Study Builder
 - a. Select **Glasses** (study type) and **Glasses live recording** (Glasses study type)
 - b. For Sensors, select **Neon Glasses** for **Eye tracking**.
2. Add a Respondent to the study and click **Start Recording**. The Neon Companion App will display the recording status (as shown in the image below).
3. Click on **End Recording in iMotions** when data collection is complete - this will stop the recording in the Neon Companion App.



Collecting Data for Import Glasses Studies/Glasses Sync

For Import Glasses Studies and Glasses Sync Studies - Neon recordings must be started and stopped in Neon Companion App.

After the opening the Import window (as described below) - the same files, file transfer steps, and import folder structure then apply (regardless of study type).

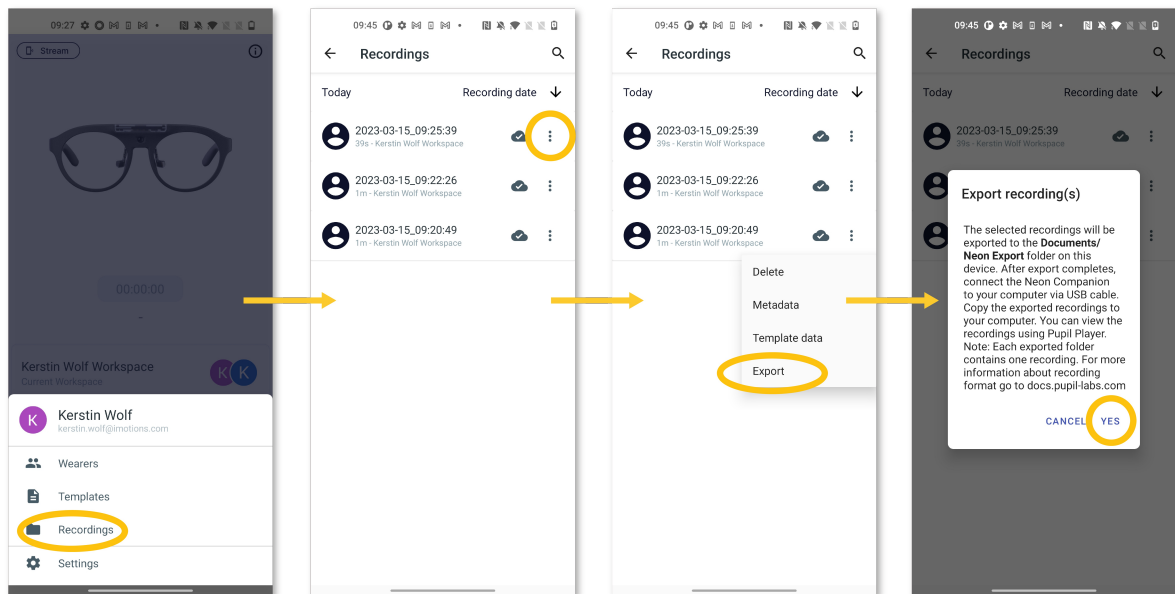
Transferring Neon Glasses Files to Computer

Neon Glasses recordings can be imported from the Neon phone directly via USB or via upload/download from Pupil Cloud website.

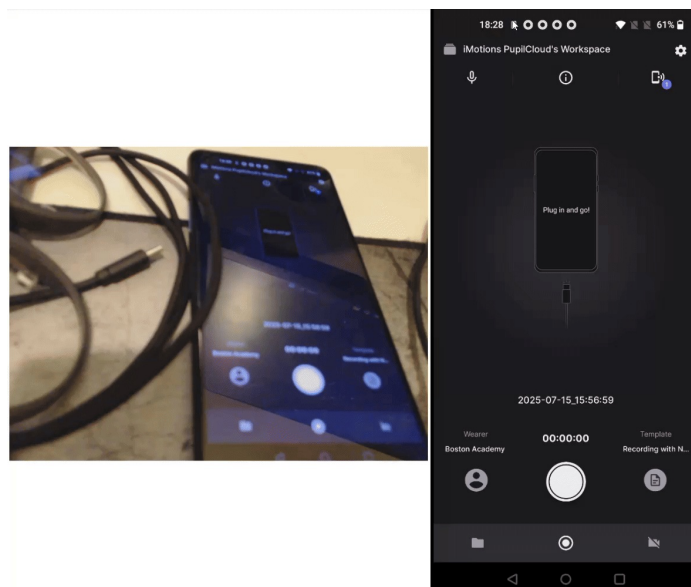
Both options allow import of the scene camera recording with 200 Hz gaze data and other available data (in iMotions 10.1.13 or newer). Additional data may be available with Pupil Cloud files and may not be required for some data in older version of iMotions (e.g. blink, pupil, accelerometer data in iMotions 10.1.12 or older).

Transfer Files via USB Cable (recommended)

1. In the Neon Companion App, open the **Recordings** menu (folder icon at the bottom of the app).
2. Open the menu for the recording (... icon to the right of the recording) and click **Export**.
3. Connect the phone to your iMotions computer with the provided USB-C cable (other USB cables may not support transfer).
4. Use Windows File Explorer to open the OnePlus storage folder: "Internal shared storage">"Documents">"Neon Export".
5. Transfer recorded files onto your iMotions computer and into a Data Folder that contains a sub-folder for each respondent and their Neon Glasses files.



Neon Companion App Export Steps



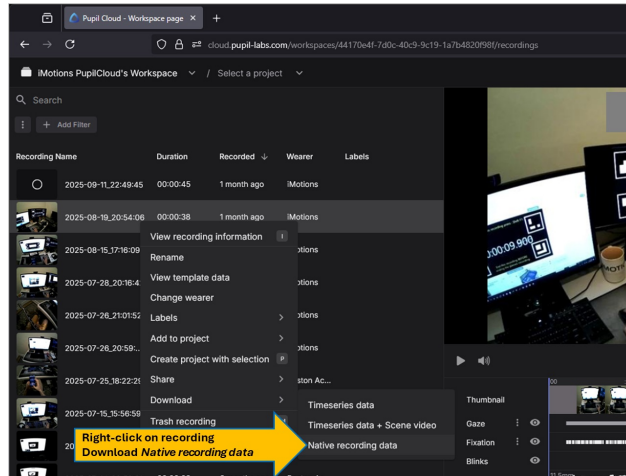
Neon Companion App Export Steps (video)

Transfer Files via Pupil Cloud (alternative)

Pupil Cloud is entirely optional for iMotions 10.1.14/Neon Companion 2.9.3 or newer.

Pupil Cloud processing is required for older versions of iMotions and Neon Companion App (described in *Requirements* and *Signals* sections, above). Eye tracking data and video are available without Pupil Cloud (regardless of software versions).

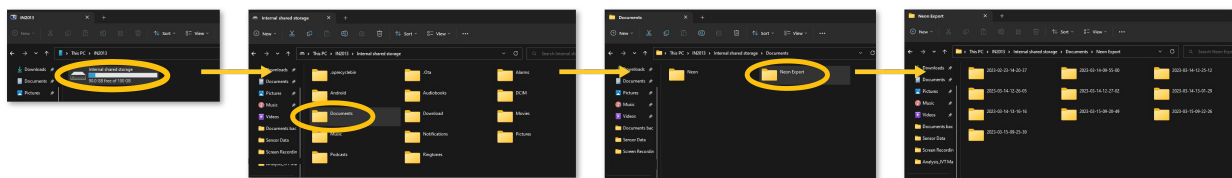
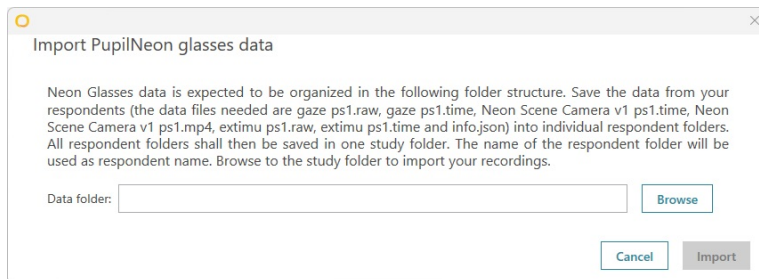
1. In the Companion App, **Cloud upload** must be enabled in **Settings** (gear icon at top right of the app).
2. Open the **Recordings** menu (folder icon at the bottom of the app).
3. Open the menu for the recording to upload (... icon to the right of the recording) and click **Upload**.
4. The upload status will appear followed by a cloud icon, when the recording is finished uploading to Pupil Cloud.
5. (Optional recommendation) Allow Pupil Cloud time for data processing to ensure expected data are available for download. Processing times are generally the same as recording durations, but will vary Pupil Cloud bandwidth.
6. **Download files** from Pupil Cloud.
 - a. Log in to **Pupil Cloud** and select the workspace and project associated with your recording(s).
Note: We recommend adjusting your recording name(s) here, before you downloading.
 - b. Select the recording(s) you want to import into iMotions from the list (Use ctrl+click or shift+click to select multiple files).
 - c. Download them by right-clicking on a selected recording, and then hovering the mouse over "Download", and then clicking on **'Native recording data'**.
This will create a zip file of your downloaded recordings carrying the name of the recording as displayed in the web interface appended with an identifier.



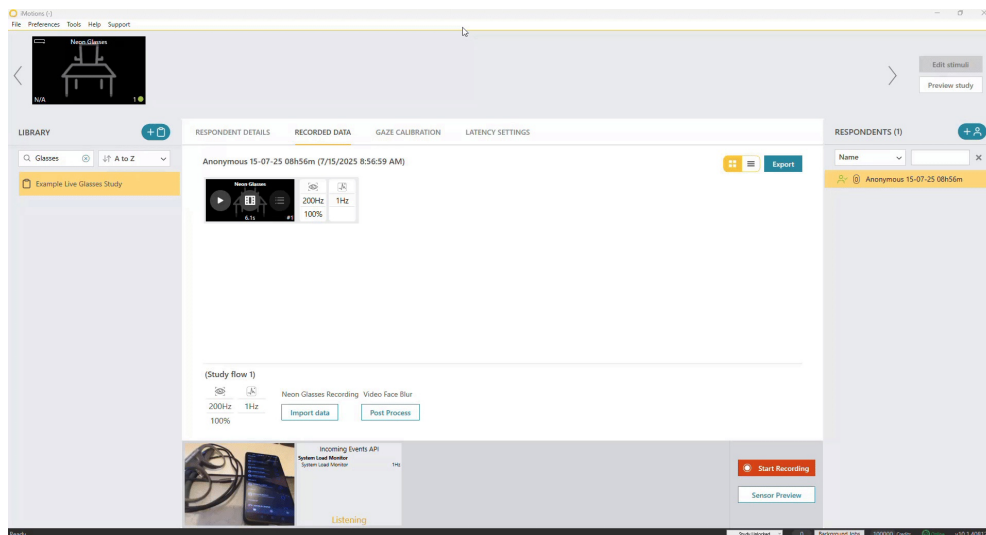
Transfer Files to Import Folder

Neon Glasses files must be transferred into a Data Folder for iMotions - as described in a related article: [Data Folder and Files for iMotions](#). The Import windows will also provide information about the Import Folder and sub-folders containing the required files:

- gaze ps1.raw
- gaze ps1.time
- Neon Scene Camera v1 ps1.time
- Neon Scene Camera v1 ps1.mp4
- extimu ps1.raw
- extimu ps1.time
- info.json



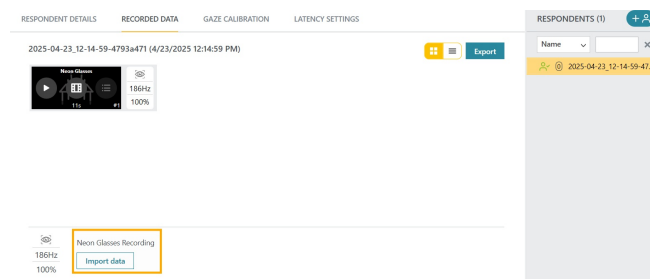
Export Folder Neon Glasses Files



Importing Neon Files into iMotions

Import into Live Glasses studies

1. Select a Live Glasses study and Respondent with Neon Glasses recording.
2. In the Recorded Data panel, click on the "Import data" button.
3. Select your Import Folder (described above).



Import Glasses Studies and Glasses Sync

Import Glasses Studies will import and only contain Neon Glasses recording/data.

Import Glasses Studies can then be merged into Glasses Sync studies.

To initiate import for Import Glasses studies:

1. Create a study with **iMotions Study Builder**
2. Select **Glasses** (study type) and **Import glasses data** (Glasses study type)
3. For Sensors, select **Neon Glasses** for **Eye tracking**.
4. Choose between "New study" or "Merge into existing compatible study".
5. Click Next to proceed to the Import window.
6. Click on **Browse** and select the folder with your Neon Glasses files, ensuring that the selected folder is set up as described in the import window.
7. Select your Import Folder (described above).
8. To merge with a Glasses Sync study, right-click on the new/imported study and select **Merge with another study** (additional information: Glasses Sync).

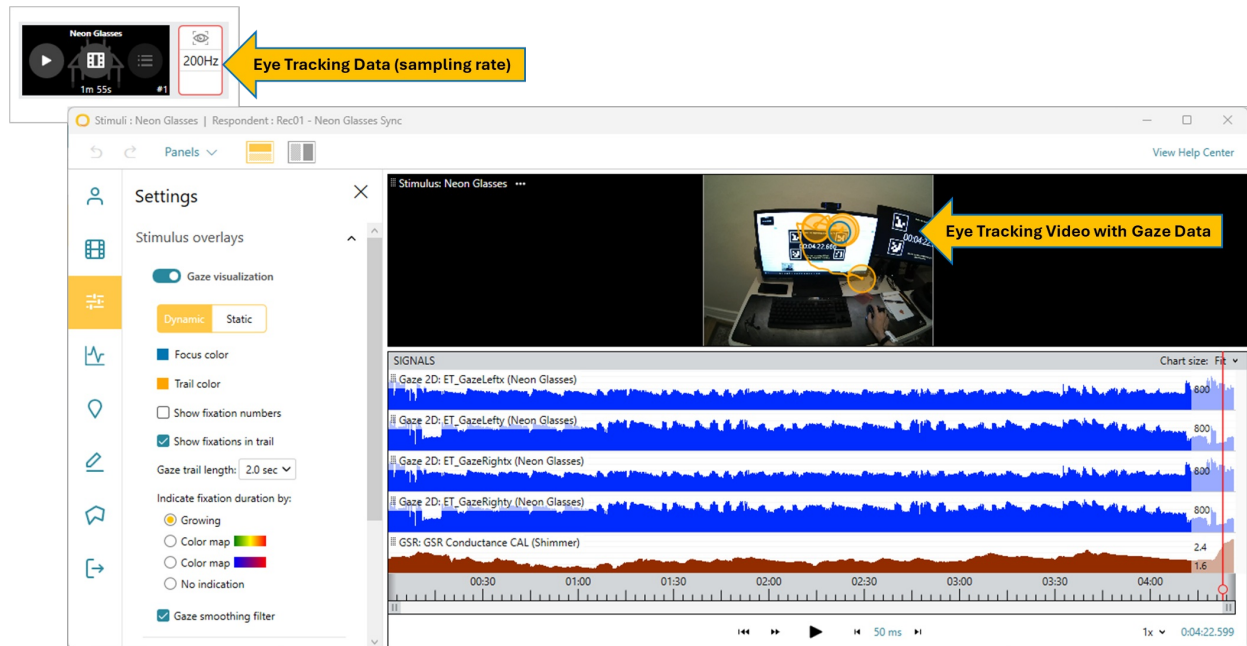
Note

It is possible, in iMotions, to remove the **fish-eye lens distortion** that is present in Pupil Labs recordings. This creates an undistorted view of the recording, which aside from the aesthetic benefits, can also aid the accuracy of **gaze mapping** processing. Please note however, that this is a resource-intensive process, and will in most cases take as long to process as the recording itself.

Imported Neon Recordings

Once Neon Glasses recordings have been imported for a Respondent:

- The Recorded Data panel will show the sampling rate for imported Neon Glasses recordings.
- The **Replay Window** will show the recorded video and gaze data, along with any other enabled/recorded data.



Latency Settings

Latency Settings can be used to adjust for unexpected latency (or lag) between the Glasses video and data.

This is most relevant for Live Recordings due to the involvement of wireless network connections (which start and end the recording). iMotions estimates and corrects for this lag, but may be inaccurate. In these cases, the latency must be visually-estimated then adjusted in the Latency Settings per respondent.